INTRODUCTORY SOMMELIER COURSE AND EXAM

AND

CERTIFIED SOMMELIER THEORY EXAM

CURRICULUM GUIDE

PURPOSE OF THIS DOCUMENT

The purpose of this guide is to outline the curriculum and general scope of subjects covered and examined to at both the Court of Master Sommeliers, Americas Introductory Sommelier Course and Exam and the Theory portion of the Certified Sommelier Exam.

This guide is an evolving document that will be updated regularly as laws, priorities, and relevance change region by region.

Please note the topic framework for each subject and the progression of knowledge between the two levels. The Certified Sommelier Exam curriculum includes and builds from the information listed in the Introductory Sommelier curriculum. A blank topic category for a particular subject indicates that information is not essential at that level.

The Certified Sommelier Theory Exam is not exclusively written from this curriculum. This guide provides direction and guidance to focus and strategize study. Success at the Certified Sommelier Exam requires additional research, supplemental resources, and work experience.

Please see the additional resources provided by the CMS-A for your preparation at: https://www.mastersommeliers.org/resources

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INTRODUCTORY & CERTIFIED THEORY CURRICULUM

Subject Name: GRAPE GROWING - VITICULTURE

Learning Outcome Introductory Sommelier Course: After studying this section, a student will understand the important factors that influence grape growing (viticulture) that impact the final flavor of the wine.

Learning Outcome Certified Sommelier Exam: After studying this section, a candidate will understand more detail of grape growing (viticulture), including the vine cycle, parts of the vine, and important viticultural terms.

	Introductory Level	Certified Sommelier Exam
Definition of Viticulture	Factors of production that influence grape growing. Factors occurring from nature plus manmade decisions in the vineyard that affect wine quality.	**Common types of viticulture:** Organic Biodynamic Sustainable Dry farming
Factor 1: Location and Geography	**Location and Geography Considerations:** Site location of vineyard Goals of wine producer Regional wine laws Financial considerations	
Factor 2: Climate - Terms	**Climate General Definition:** The weather conditions in an area in general or over a long period of time. This includes: temperature, humidity, air pressure, rainfall, sunshine, cloud cover, wind, and other elements that are relatively stable from year to year. **Climate and weather are different!* **Climate types:** Continental Maritime Mediterranean Desert/Arid **Climate descriptors:** Cool Moderate Warm Hot	**Macroclimate:** the climate of an area, such as a wine region **Mesoclimate:** the climate of a vineyard **Microclimate:** the climate around a single vine **Degree-Days:** Days during the growing season are assigned individual growing degree-days according to the number of days that the average temperature exceeds the threshold of over 50°F.
Factor 2: Climate - Latitude	**Where do grapes grow?** **Latitudinal Wine Growing Zones:** Most wine-growing regions are within the 30-50° degree latitudinal lines. Wine-growing regions may exist outside this range mainly due to climate moderators.	
Factor 2: Climate - Moderators	**Climate Moderators:** Factors that affect the general climate of an area such as: Altitude Diurnal shifts Sunshine hours Bodies of water Wind Rain shadow effect	Understand how major wine regions of the world experience climate moderation and how the moderation affects grape growing. Examples include: Alsace, Rhône Valley, Chile, Mendoza, parts of Napa and Sonoma County, California, Columbia Valley, Washington and many (others). Examples of moderators include the San Pablo Bay, Benguela Current, Humboldt Current, etc.

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Factor 3: Topography	The surface features of the land, such as local elevation or altitude. This includes local mountains, hills, valleys, and bodies of water.	Know topography for important wine regions and how it affects grape growing. Examples include: Vosges Mountain Range, Cascade Mountain Range, Andes Mountains... The important features are listed in the curriculum for that region if it's relevant at this level.
Factor 4: Soil	Properties of different soil types provide specific things to a vine's health and growth. Soil types are matched with grape varieties based on the climate and needs of the grape variety. Some grape varieties do best in certain soil types. **Soil properties:** Drainage or water retention Sun reflection Nutrient content Temperature regulation Pest resistance **Soil type examples:** Clay, slate, limestone, granite, chalk, sand, shist Be able to match soils to these major regions: Champagne, Chablis, Côte d'Or, Haut-Médoc, Pomerol, St.-Émilion, Châteauneuf-du-Pape, Mosel. The important soils to note are listed in the curriculum for that region if it's relevant at this level.	Match the soils types to a more extensive list of wine regions such as: Champagne, Chablis, Côte d'Or, Coonawarra, Gimblett Gravels, Haut-Médoc, Pomerol, St.-Émilion, Châteauneuf-du-Pape, Cornas, Hermitage, Côte-Rôtie. The important soils to note are listed in the curriculum for that region if it's relevant at this level.
Factor 5: Grape Variety	**Genus:** Grapes are the fruit of a vining plant of the genus **Vitis.** **Species:** There are many species of vining plants that produce grapes for making wine. **Vinifera** is the focus here. **Vitis vinifera:** Native to Mediterranean, Europe, Southwestern Asia. Most common for quality grape production. 10,000 varieties. Examples: Cabernet Sauvignon, Merlot, Chardonnay, Riesling, Syrah **How are grape varieties chosen?** Regional wine law Climate compatibility Soil compatibility Marketability	**Parts of the grape vine:** Rootstock Trunk Cordons Shoots/canes Grape clusters/bunches Canopy **Other viticulture terms:** Hybrid Crossing Clone Mutations **Other species of vines:** **Vitis Labrusca** examples: Catawba, Concord
Factor 6: Vintage - Yearly Weather	Grape growing or viticulture is farming. Every year produces a different crop called the vintage. Each vintage can be different and an indicator of style, quality, and price potential. Weather connection to quality. **Introductory students do no need to memorize differences in vintages at this level.**	Understanding the differences between vintages of major wine regions over the last 10-15 years. Those required for the Certified Sommelier Exam are listed in the curriculum for that region if it's relevant at this level.

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Factor 7: Vineyard Management	**Vineyard management:** Why manage a vineyard? Vineyards need to be managed to remain viable and healthy. Vines need to be managed or trained so they vine in a particular size and ideal shape to ripen grapes best. Many vine training systems exist to suit different grape varieties in wine-growing regions. **Canopy management:** Managing the growth of the leaves. **Irrigation:** Making sure the vine has enough water in the growing season. Managing pests, disease, mildew, fungus infections, etc.: **Fungi/Mold:** Botrytis cinerea *(can be a good thing!)* **Pests:** Phylloxera **Managing yield:** Making sure the vine is producing the amount of fruit desired. **Green harvest:** a quality measure of removing or dropping unripe grapes or clusters.	**Vine Cycle:** Dormancy Bud Break Flowering Fruit Set Ripening - Veraison - Hangtime Maturity Harvest Pruning **Vineyard Architecture:** Vineyard slope and sun exposure **Vineyard aspect:** Resulting characteristics of finished wines that are exposed to heat and light at different times of day. **Vine Training Systems:** **Two main systems to note.** 1. Head training: Examples such as Gobelet/bush 2. Cordon training: Using support for growth
Factor 7: Vineyard Management - Pests and Disease	**Managing pests and disease:** **Phylloxera:** Microscopic vine louse Mid-19th century destroyed many of the vineyards in France. Use of resistant American vine species rootstock to stop the spread. Most wine regions of the world are affected.	**Phylloxera:** Grafting of Vitis Vinifera onto Vitis Labrusca rootstock. **Other vine diseases:** Pierce's Disease
Factor 8: Yield	**Yield:** How much fruit a vine produces. This is influenced by many factors and controlled (or not) by human intervention. Yield can be measured in a few ways: - Tons per acre - Hectoliters per hectare	Understand conversion formulas to navigate between metric and standard systems. Understand the quality impact as a result of yield choices and how different grape varieties respond to various yields.

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Factor 9: Harvest	**Timing of picking -** Considerations include: Weather Style of wine being made Producer goals Regional wine law Number of vineyard passes for picking **How grapes are picked?** By hand or by machine **How are the grapes collected?** Collection bin size (large or small) ** How does this affect quality?* **Grape maturity:** Ripeness is a range, one vineyard manager's definition of ripeness may be different than another's. Sugar measurements: Example- Brix Structure and flavor ripeness	Night harvests Phenolic ripeness versus sugar ripeness. How do choices made in the vineyard affect phenolic vs sugar ripeness
Factor 10: Terroir	**"Goût de terroir" or Taste of Earth:** Encompasses all of the environmental impacts on a grapevine. Unique characteristics of wine such as aromas and flavors that distinguish it from all others.	
Factor 11: Regional Wine Law	Regional wine laws dictate the grape varieties can be grown and how they are grown in many wine regions of the world.	

INTRODUCTORY & CERTIFIED THEORY CURRICULUM

Subject Name: WINEMAKING - VINIFICATION

Learning Outcome Introductory Sommelier Course: After studying this section, a student will understand the definition of winemaking (vinification), the steps of making white, red, and rosé, and other important factors of consideration that influence the how and why wines taste the way they do.

Learning Outcome Certified Sommelier Exam: After studying this section, a candidate will have a deeper understanding of winemaking cause and effect on the final wine aromas and flavors, such as fermentation techniques, barrel size choices, types of oak used, and other important winemaking decisions.

	Introductory Level	Certified Sommelier Exam
Definition of Winemaking - Vinification	**Vinification:** Decisions and practices made in the winery from harvest to when the wine is bottled.	Basic chemistry of wine and how it relates to cause and effect. Malolactic conversion, acids in wine, and sugars in wine.
Factors in Decision Making	What is the goal of the winery? What style of wine will be produced? What will the price point of the wine be? Are there regional wine laws the winery must adhere to? Sustainability.	Breakdown of how sustainability is implemented in the winery.
Winemaking Steps	**Harvest** **Sort** **Destem** **White Wines:** Crush, Press, Fermentation, Fining/Filtering, Bottling/Packaging. **Red Wines** Crush, Maceration, Fermentation (with skin contact), Pressing, Aging, Fining/Filtration, Bottling/Packaging, Bottle Aging. **Rosé Wines (simple version)** Crush *(red grape varieties),* Maceration skin contact *(for a short time),* Pressing, Fermentation, Fining/Filtration, Bottling/Packaging.	**Three types of Rosé production:** Maceration time Blending Method - often used for sparkling wine production. Saignée* *Saignée or "bleeding" method (this essentially doesn't exist anymore. It was a concentration technique for red wine production with rosé being the byproduct.) **Aging:** Can be on the lees or without. Can be in concrete, stainless steel, or oak.
Alcoholic Fermentation	Sugar + Yeast --> Alcohol, CO_2, Heat, Aromas, Flavors	Saccharomyces cerevisiae = species of yeast for wine fermentation
Fermentation Vessels Types	**Type of vessel** Wood Stainless steel Concrete	Exploration of oak and its various origins. Identification of other types of woods used for barrels, such as acacia, cherry, and chestnut. **Barrel Sizes:** Barrique = 225 liters Puncheon = 300 liters Demi-muid = 600 liters Foudre/Botti = 1,000+ liters Understand the cause and effect of the different types of vessels and which has more flavor impact. Know which major areas of the world use them, such as botti in Piemonte or barriques in Rioja.
Fermentation Vessels Types and Considerations	**Size:** Large or small. **Shape**: Oval, round, egg. Oxygen permeability considerations. Temperature control.	Basic knowledge of coopers and the construction of barrels.

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	Introductory Level	Certified Sommelier Exam
Winemaking Terms	Grape must Chaptalization Acidification Water addition Lees Autolysis Malolactic fermentation Carbonic maceration	**Cause & effect:** Skin contact and its effect on color, extraction, pH, and texture. Bâttonage and its effect on weight, flavor, and texture. Whole-cluster versus stem inclusion versus whole-berry fermentations. Be able to associate winemaking practices to a non-fruit descriptor on the Certified Exam Tasting Grid. Know which regions and grape varieties are associated with them and which are not: lees/autolysis, malolactic fermentation, carbonic maceration.
Winemaking Decisions	Temperature control and why it's important for different color wines and styles. Oxygen management and its impact of final style of wine produced. Extraction explained through the use of punch downs, pump overs, maceration times, and fermentation temperature.	Outlining the differences between reductive and oxidative winemaking.
Aging and Maturation Considerations	Costs Age of barrels Length of time of aging	What each region dictates and what options winemakers have.
Preparation for Bottling	Clarification and the difference between fining and filtration.	Cold stabilization Examples of fining agents Sulphur additions and the difference between free and bound compounds.
Packaging	**Containers:** Bottles Kegs Boxed wine **Closures:** Cork Composites Screw cap (Stelvin) Glass top (Vinolok) **Packaging:** Labels and label design Capsule colors and design Adhering to governmental label laws	